College Board AP® Chemistry

Your notes

Periodic Trends

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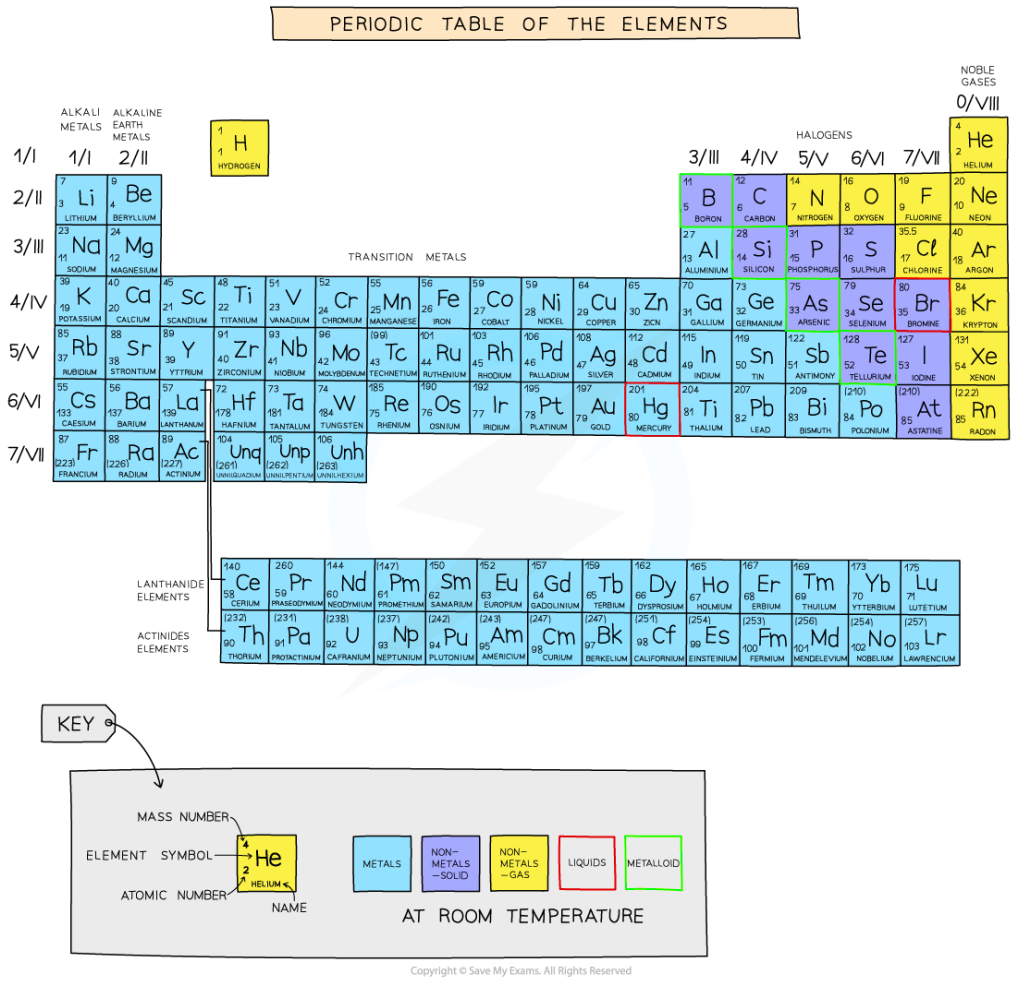
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Your notes

Structure of the Periodic Table

- The **periodic table** is a tabular arrangement of chemical elements organized based on their **increasing atomic number,** from 1 to 118
- The periodic table is organized into horizontal rows called **periods** and vertical columns called **groups**

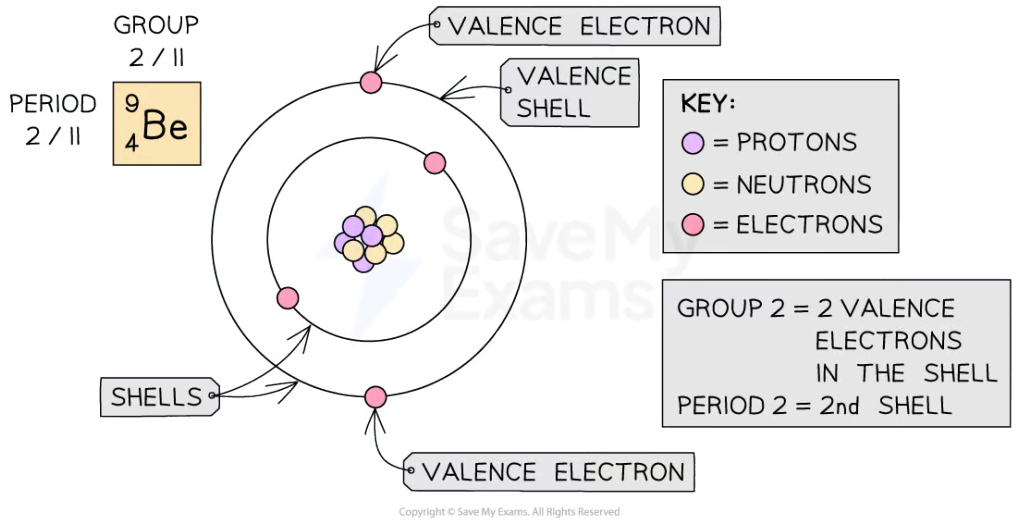
The Periodic Table



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- **Periods** represent the principal **energy shells** of an element
- **Groups** are columns that share **similar chemical properties** due to the number of **valence electrons**

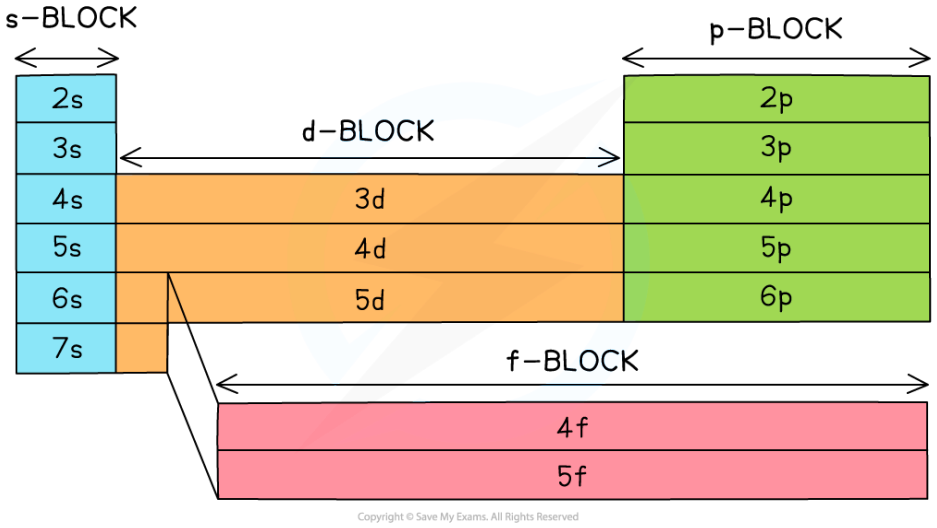
The Structure of the Beryllium Atom



The beryllium atom in the periodic table

- The periodic table is divided into four block types: **s-block**, **p-block**, **d-block**, and **f-block**
  - **s-block elements** have only s electrons in their valence shell
  - **p-block elements** have at least one p-electron in their valence shell
  - **d-block elements** have at least one d-electron and one s-electron, but no p or f electrons in their outer shell (up to 5d)
  - **f-block elements** have at least one f-electron and one s-electron, but not p or f electron in their outer shell

Blocks in the Periodic Table



s-block, p-block, d-block, and f-blocks in the Periodic Table

Your notes

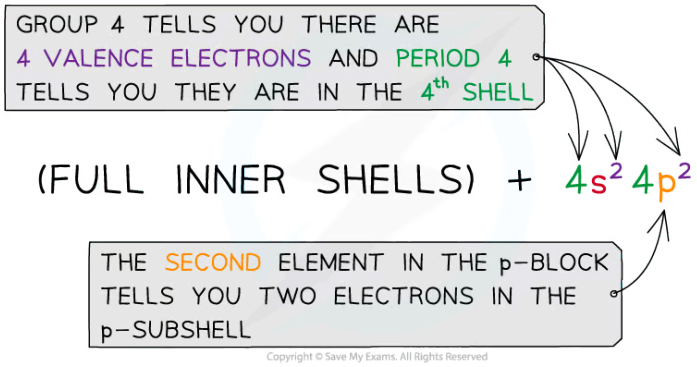
Electron Configurations



Your notes

- The arrangement of electrons in the energy shells and subshells of an atom is described by its electron configuration
- The electron configuration of an element can be predicted by using the periodic table
- The image below shows how this can be done for the germanium atom:

Electron Configuration of the Germanium Atom



Deducing the electron configuration of germanium as $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3d^{10}$, $4s^2$, $4p^2$

- Germanium is in the Group 4 (skipping the transition metals) and in Period 4
- Group 4 tells you that it has 4 valence electrons
- Period 4 tells you that it has the 4 valence electrons in the fourth shell
- Finally, germanium is in the second position of the p-block of the periodic table, therefore, two electrons must be in the **p subshell**

Your notes

Periodic Trends

- **Periodicity** refers to the recurring and predictable trends in the properties of elements as you move across the periodic table
- These trends can be understood through principles such as **Coulomb's law**, the **shell model**, and the concept of **shielding/effective nuclear charge**

Ionization Energy

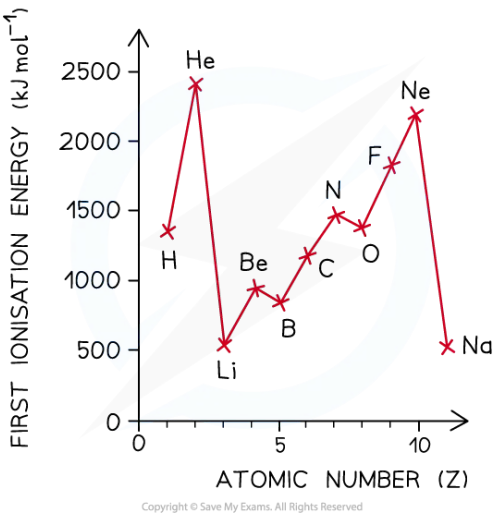
- **Ionization energy** is the energy required to remove one mole of electrons in gaseous state from one mole of neutral gaseous atoms, forming one mole of positively charged gaseous ions
  - e.g. The equation below shows the first ionization energy of sodium (Na)

$$Na_{(g)} \rightarrow Na^{+}_{(g)} + e^{-}$$

- As you move across a period (from **left to right**), the ionization energy generally **increases**
  - There is an **increase in the effective nuclear charge** because the number of protons increase
  - Therefore, the **coulombic attraction** between the nucleus and the valence electrons is **stronger,** so it is **harder to remove an electron**
  - The **shielding does not play an important role** along the periods, since atoms in the same group have the same shielding effect because the **number of inner shells is the same**
  - The **distance** between the nucleus and the valence electrons is reasonably the **same**

A graph showing the ionization energies of the elements hydrogen to sodium

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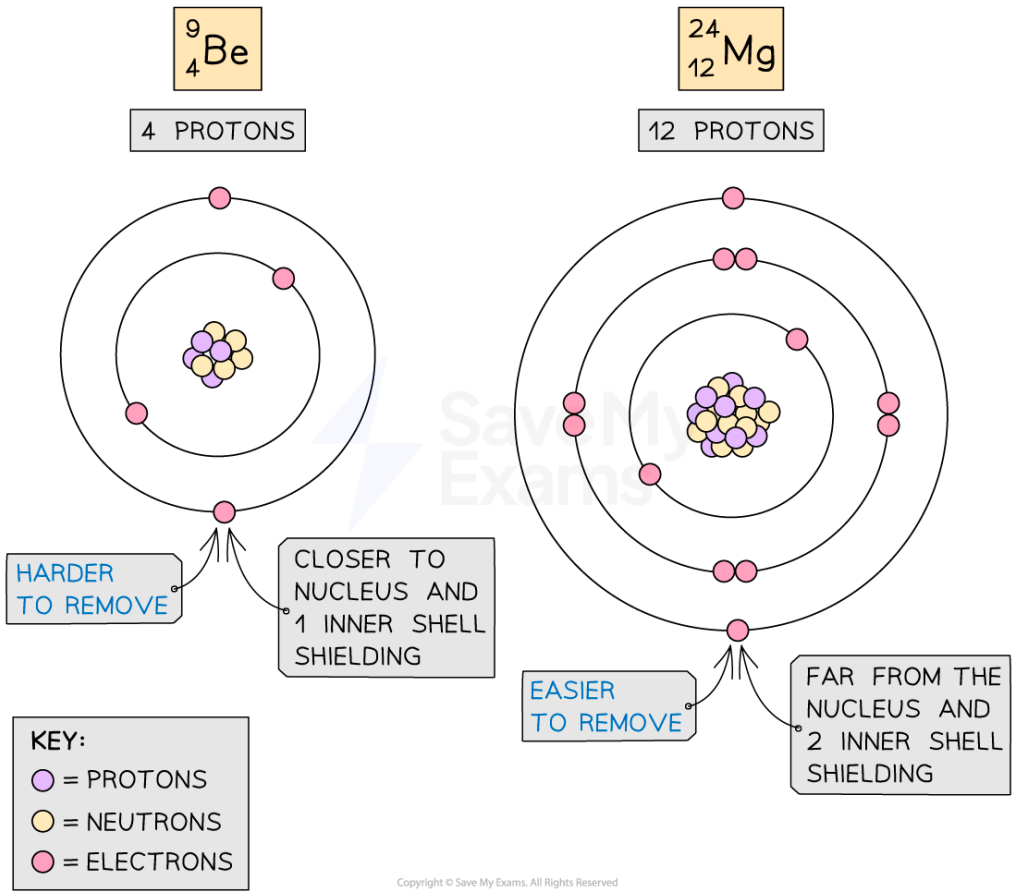
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There is a general increase in ionization energy from left to right

- As you move down a group (from **top to bottom**), the ionization energy **decreases**
  - This occurs because the **shielding effect increases** because the **number of inner shells increases**
  - The **distance** between the nucleus and the valence electrons increases, making the **Coulombic attraction considerably weaker**
  - Therefore, the **effective nuclear charge decreases** and its easier to remove an electron

A Beryllium and Magnesium Atom



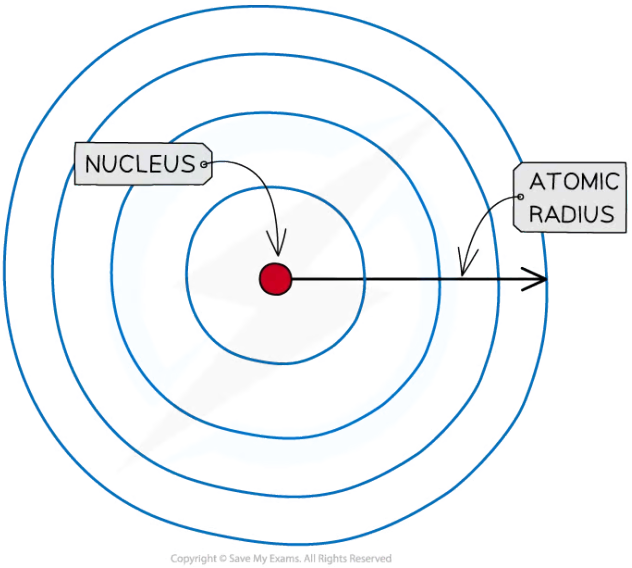
Comparison, between beryllium and magnesium, of the factors that affect the ionization energy

Atomic Radius

- **Atomic radius** is the size of an atom
  - It can also be defined as the distance from the nucleus to the outermost electron

Atomic Radius

Your notes

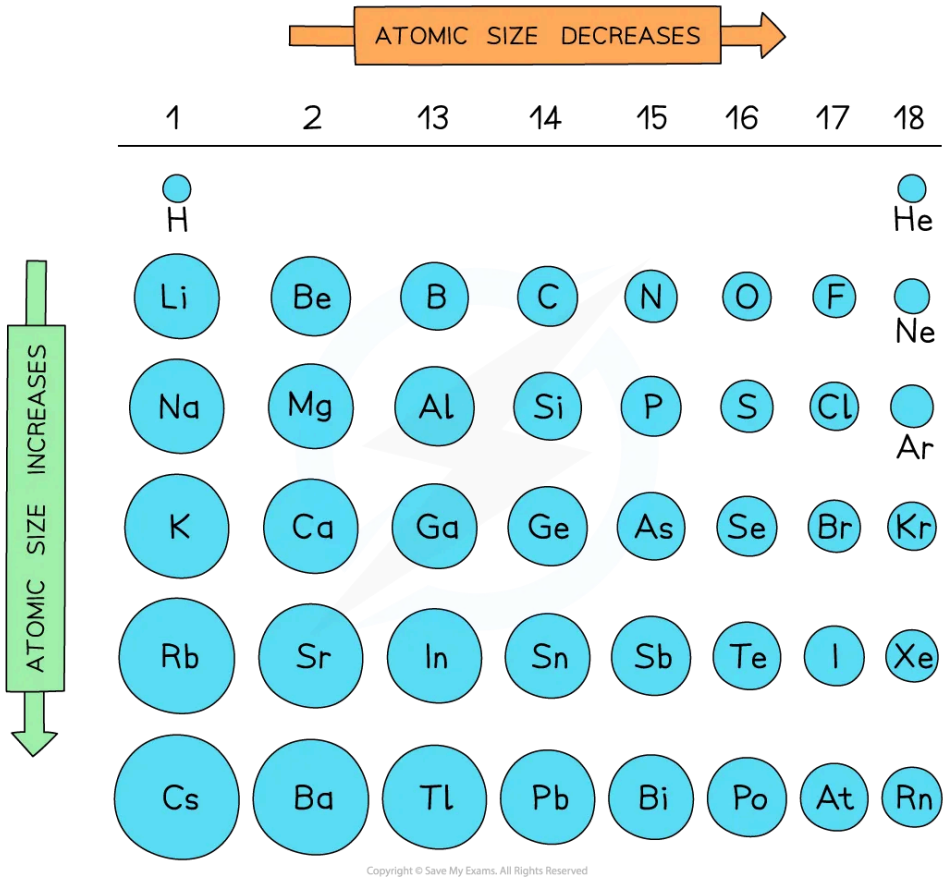


The atomic radius of an atom is the typical distance between the nucleus and the outermost electron shell

- Atomic radius and ionization energy have opposite trends
- As you move across a period (from **left to right**), the atomic radius generally **decreases**
  - There is an **increase in the effective nuclear charge** because the number of protons increase
  - Therefore, the **Coulombic attraction** between the nucleus and the valence electrons is **stronger,** which results in smaller atoms
  - The **shielding does not play an important role** along the periods, since atoms in the same group have the same shielding effect because the **number of inner shells is the same**
- As you move down a group, the atomic radius **increases**
  - This occurs because the **shielding effect increases** because the **number of inner shells increases**
  - The **distance** between the nucleus and the valence electrons increases, making the **Coulombic attraction considerably weaker**
  - Therefore, the **effective nuclear charge decreases** and the valence electrons are not pulled strongly, meaning bigger atoms

Atomic Radii in the Periodic Table

Your notes



Trends in the atomic radii across a period and down a group

Ionic Radius

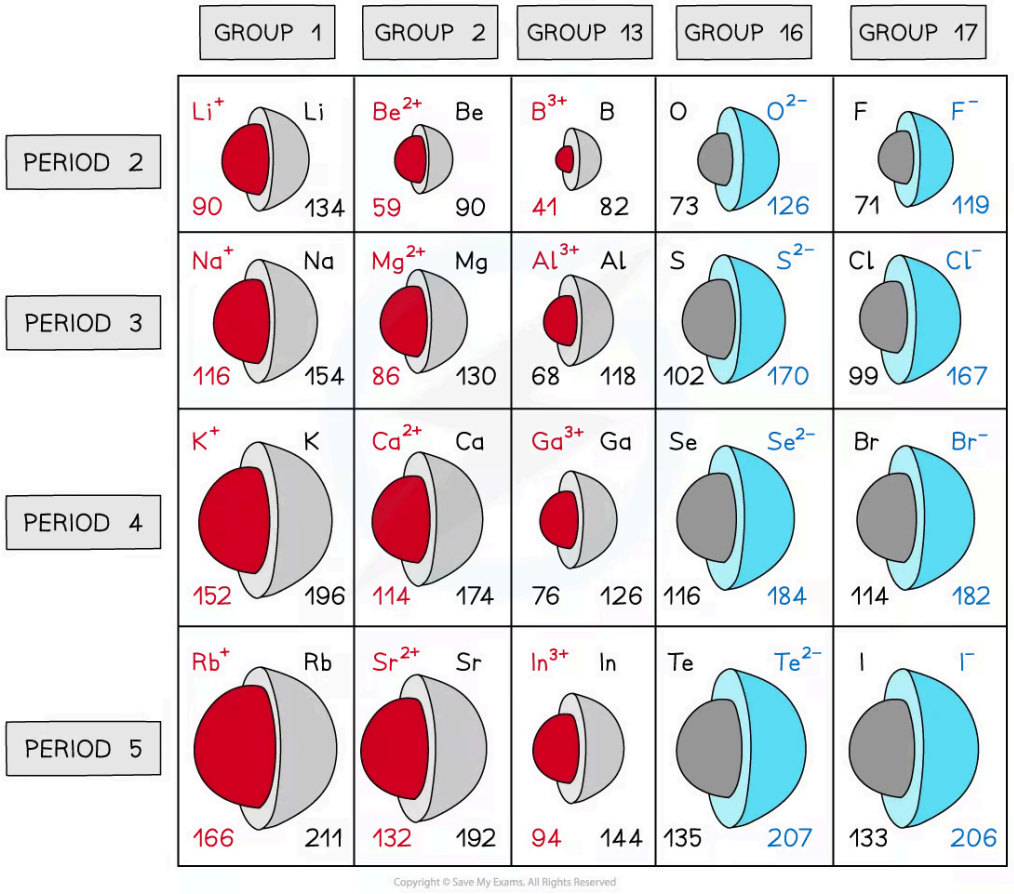
- **Ionic radius** is the size of an ion, which can be larger or smaller than the corresponding neutral atom
- Down a group it follows the same pattern as atomic radius
  - Ionic radii increase as energy shells are added, **decreasing the effective nuclear charge and the Coulombic attraction from the nucleus**
- However, there is a different periodic trend for **positive ions (cations)** and **negative ions (anions)**
- The ionic radius **increase** with an increasing negative charge
  - Negative ions are formed when electrons are accepted, while the nuclear charge is the same
  - Adding electrons creates extra repulsion with the other valence electrons, which means a bigger ionic radius
  - The **greatest the negative charge, the largest the ionic radius**
- The ionic radius **decreases** with an increasing positive charge

Your notes

- Positive ions are formed when electrons are lost, while the nuclear charge is the same
- Removing electrons decreases the repulsion within the valence electrons, which means a smaller ionic radius
- The **greatest the positive charge, the smallest the ionic radius**

Your notes

Sizes of Atoms and their Ions in pm



Trends in the ionic radii across a period and down a group

Electron Affinity

- **Electron affinity** is the energy change when one mole of electrons are added to one mole of a neutral atom in gaseous state, forming one mole of negatively charged ions
  - E.g. The equation below shows the first ionization energy of sodium (Na)

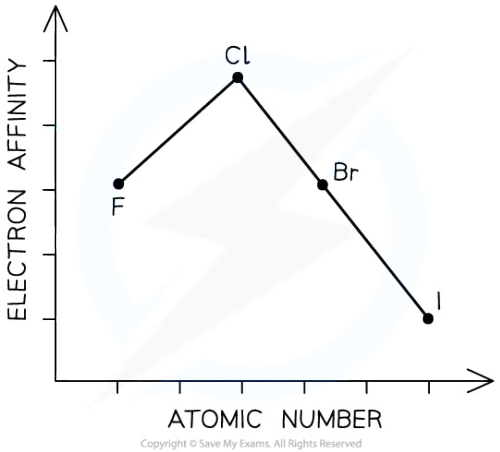
$$Cl_{(g)} + e^- \rightarrow Cl^-_{(g)}$$

- Electron affinity is an **exothermic process,** this means that energy is released
  - When a process release energy, the sign of the energy is **negative**
- Elements on the Group 7 (F, Cl, Br, I) of the periodic table have the most negative electron affinities

- This occurs because these elements tend to be stable as ions with a -1 charge, by completing the p subshell
- Electron affinity generally decreases down the group
  - This occurs because the **shielding effect increases** because the **number of inner shells increases**
  - The **distance** between the nucleus and the valence electrons increases, making the **Coulombic attraction considerably weaker**
  - Therefore, the **effective nuclear charge decreases** and the coulombic attraction to additional electrons is less
  - An exception to this rule is **fluorine** since an additional electron in the 2p subshell create a considerable repulsion with the other valence electrons

Your notes

Electron Affinities



Electron affinities down Group 17 from F to I

Electronegativity

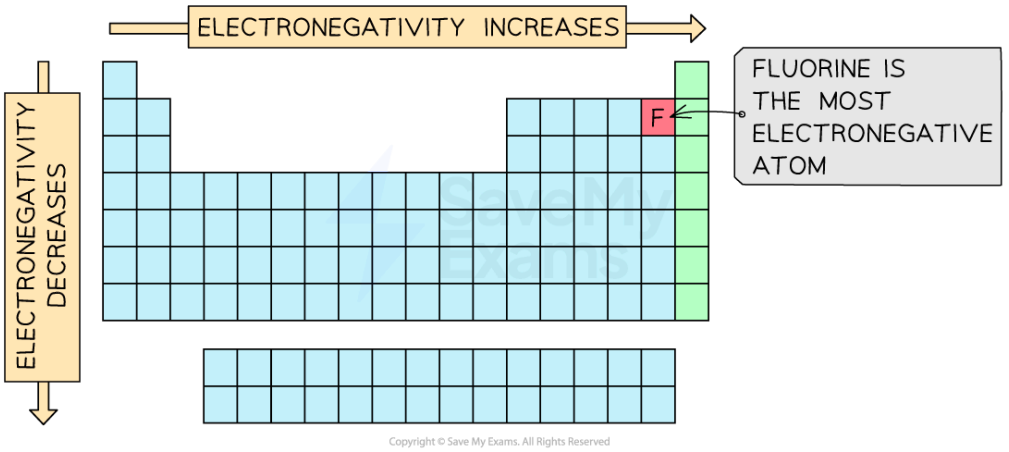
- **Electronegativity** measures the ability of an atom to attract a pair of electrons when it forms a covalent bond
- Electronegativity generally increases across a period (from left to right)
  - There is an **increase in the effective nuclear charge** because the number of protons increase
  - For that reason, the **Coulombic attraction** between the nucleus and the electrons from the covalent bond is **stronger**
  - The **shielding does not play an important role** along the periods, since atoms in the same group have the same shielding effect because the **number of inner shells is the same**
- Down a group, electronegativity decreases
  - This occurs because the **shielding effect increases** because the **number of inner shells increases**

- The **distance** between the nucleus and the valence electrons increases, making the **Coulombic attraction considerably weaker**
- Therefore, the **effective nuclear charge decreases** and the coulombic attraction to the electrons from a covalent bond decrease

- Electronegativity does not apply to noble gasses since they **do not form covalent bonds**

Your notes

Trends in Electronegativity



Periodicity of electronegativity – there is an increase from left to right and a decrease from top to bottom

Examiner Tips and Tricks

When answering free-response questions from **Section 2** regarding periodic trends, make sure to always mention: **Coulomb's Law, the shell model** (inner shells and valence shell), and the **shielding/effective nuclear charge**. This is a must if you want to be awarded with full credit for the question

Your notes

Predicting Properties of Elements

- The periodic table is a tool that can be used to predict or estimate chemical properties when information is missing.
- The predictions need to be based in the position of the element in the table



Worked Example

According to periodic trends, which of the following elements has the smallest atomic radius?

1. Potassium (K)
2. Calcium (Ca)
3. Bromine (Br)
4. Strontium (Sr)

Answer:

- The correct answer is Bromine is in period 3 at the far right side of the periodic table. Bromine's nuclear charge is greater than potassium and calcium, and it has less shielding than Strontium
- Strontium is in period 4. Therefore, the shielding effect is bigger, the coulombic attraction to the valence electrons weaker, and the atom is bigger. That is why **C** is also incorrect
- **B** and **A** are not correct since they are in the same period but at the far left. Even if the shielding effect is the same, the nuclear charge of Potassium and Calcium is smaller compared to Bromine. Therefore, the coulombic attraction from the nucleus to the valence electrons is smaller, meaning bigger atoms



Examiner Tips and Tricks

Expect to be assessed in the prediction of periodic trends in both sections of your AP Chemistry Exam. Make sure to have the chemical principles behind the properties really clear



Worked Example

Using your knowledge of atomic structure, predict if the ionization energy of Chlorine (Cl) would be greater, smaller or equal than the ionization energy of fluorine (F).

Answer:

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- Chlorine has a larger atomic size and more electron shells.
- This leads to weaker coulombic attraction between the nucleus and outer electrons
- Therefore, less energy is required to remove an electron compared to fluorine
    - Fluorine has less shielding effect and the coulombic attraction to its valence electrons are stronger



Your notes

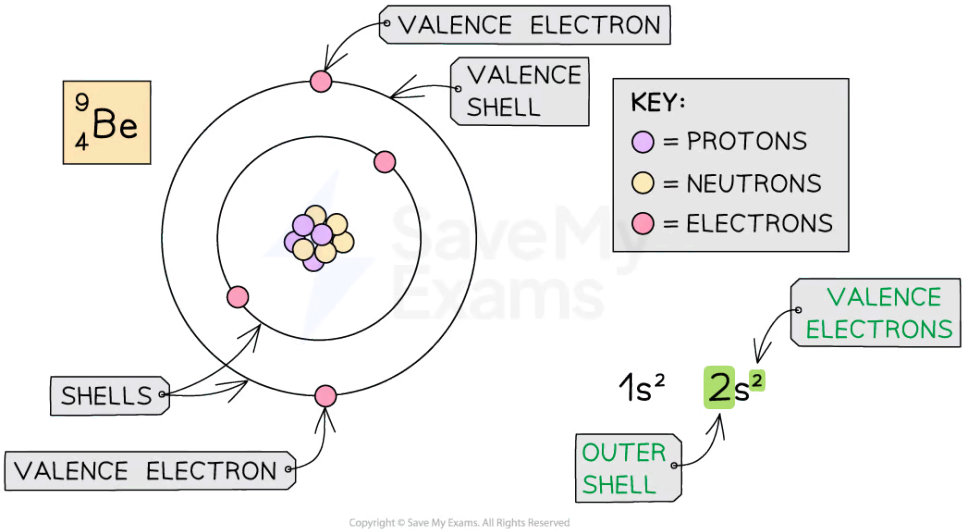
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Your notes

Valence electrons

- **Valence electrons** are the electrons present in the outermost shell of an atom
- Valence electrons dictate an atom's **reactivity** and **ability to form bonds**
- The number of valence electrons can be obtained from the **electron configuration of an atom**

The Beryllium Atom



Beryllium electron configuration and valence electrons

- The interactions between the valence electrons and nuclei of elements determine if two elements are likely to form a **chemical bond**
- There are two types of chemical bonds depending on the elements that take part in the bonding process: **ionic bond** and **covalent bond**
  - An **ionic bond** occurs between a metal and a nonmetal
  - A **covalent bond** occurs between two nonmetals

Ion Charges & Valence Electrons

Valence Electrons and the Periodic Table

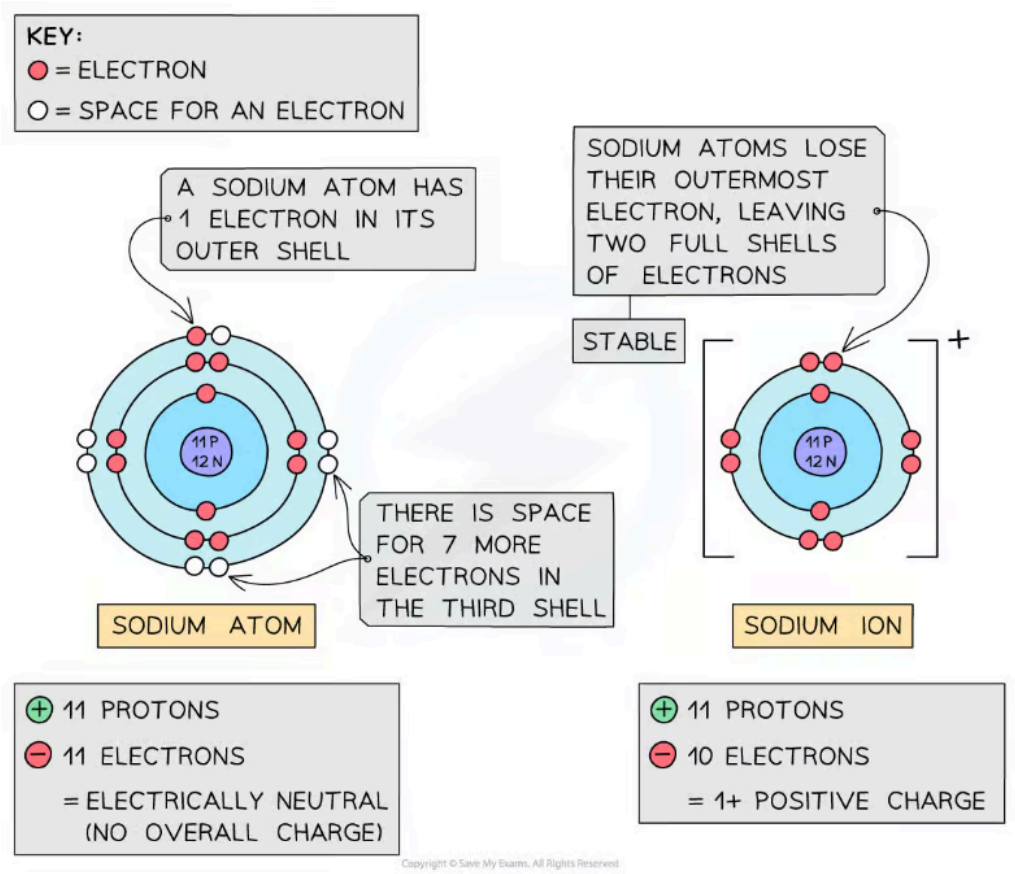
- Elements in the same **group** (columns) of the periodic table have the same number of electrons in their **valence shell**
- The valence electrons determine the chemical behavior
- As you move across a period from left to right, without counting the transition metals, the number of valence electrons increases by one

Formation of Ions

Your notes

- Ions are atoms that have lost or gained electrons
- Atoms form ions to achieve the **electron configuration of their closest noble gas**
- **Noble gasses** have a stable electron configuration because of their full outer shell
- Atoms lose electrons to form **cations**
  - Since a cation has more protons than electrons, it has a positive charge
  - **Cations** are formed by elements **metals** which have few valence electrons

Sodium Atom Becoming a Sodium Ion

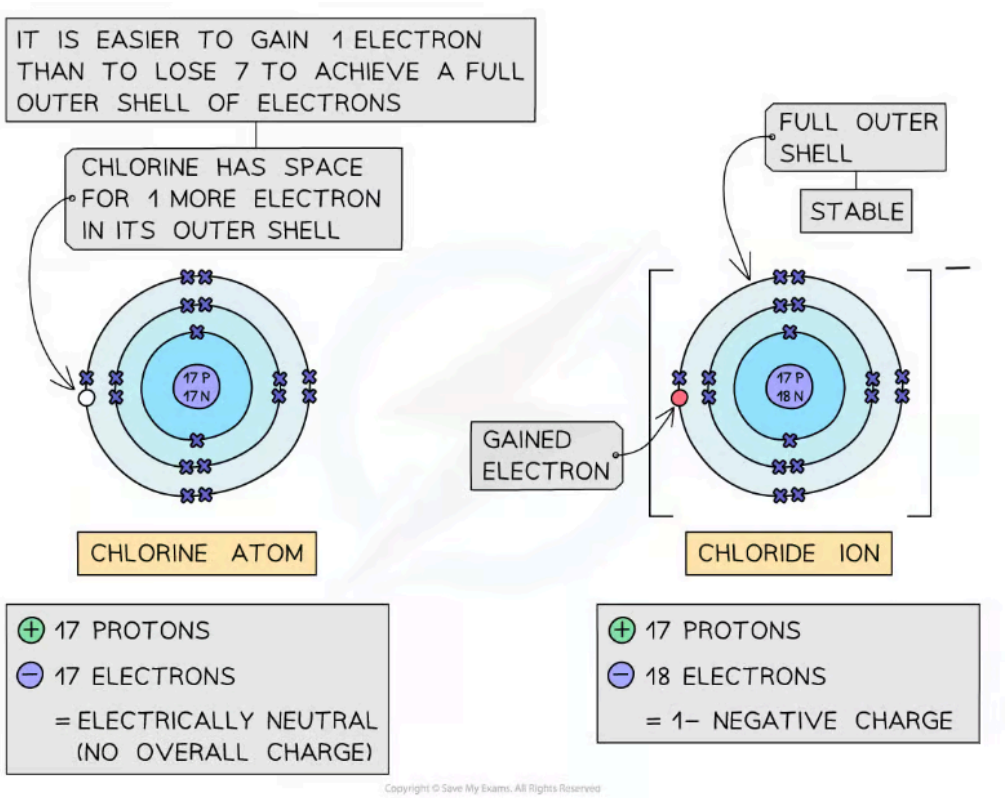


Formation of positively charged sodium ion - sodium loses 1 electron to become a sodium ion

- Atoms gain electrons to form **anions**
  - Since an anion has less protons than electrons, it has a negative charge
  - **Anions** are formed by **non-metals** which are elements that lack a few electrons
- The electrostatic attraction between a positive ion (cation) and a negative ion (anion) is defined as an **ionic bond**

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Chlorine Atom Becoming a Chloride Ion



Formation of negatively charged chloride ion – it gains 1 electron to become a chloride ion

Common ions and the Periodic Table

- The periodic table can help you to identify the typical charges of cations or ions
- Elements in **Group 1** tend to lose their lonely electron from their valence shell, therefore, the common charge for all the elements is **1+**
  - E.g. Sodium (Na) tend to form Na^+ cation
- Elements in **Group 2** tend to lose two electrons from their valence shell, therefore, the common charge for all the elements is **2+**
  - E.g. Magnesium (Mg) tend to form Mg^{2+} cation
- Elements in the **Group 3** (skipping the transition metals) tend to lose three electrons from their valence shell, therefore, the common charge for all the elements is **3+**
  - E.g. Aluminum (Al) tend to form **Al^{3+}** cation
- Elements in the **Group 5** (skipping the transition metals) tend to gain three valence electrons, therefore, the common charge for all the elements is **3−**
  - E.g. Nitrogen (N) tend to form N^{3-} anion
- Elements in the **Group 6** (skipping the transition metals) tend to gain two valence electrons, therefore, the common charge for all the elements is **2−**

Your notes

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- E.g. Oxygen (O) tend to form O^{2-} anion

- Elements in the **Group 7** (skipping the transition metals) tend to gain one valence electron, therefore, the common charge for all the elements is **1–**
  - E.g. Fluorine (F) tend to form F^{-} anion
- Atoms in **Group 4** tend not form ions, therefore it is not possible to form ionic compounds with those elements



Your notes

Examiner Tips and Tricks

From Group 1 to Group 3, the positive charge is equal to: **charge = number of group**. From Group 5 to Group 7, the negative charge is equal to: **charge = number of group – 8.** This trick works only if the groups of the transition metals are not numbered

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